ShopDesk Test Strategy

Purpose: Decide whether ShopDesk is safe to release, using repeatable evidence.

In scope

Authentication, products and search, cart, orders, stock and exact kobo totals, error responses, browser journeys.

Not implemented / out of scope

Real payment processing, vouchers/discounts in live API, free shipping in live checkout, product creation/deletion, pagination, load testing and mobile-native apps. Week 2 `web/src/lib` helpers are isolated learning exercises, not storefront capabilities.

Risk register (likelihood 1–5 × impact 1–5)

Risk	L	I	Score	Coverage
R-01 Wrong total	4	5	20	Unit, API, integration, E2E
R-02 Authentication bypass	3	5	15	API negatives, browser guards
R-03 Order without session	3	5	15	API 401 and redirect
R-04 Search wrong results	4	2	8	API query, E2E
R-05 Cart totals wrong after edits	3	4	12	Component and E2E
R-06 Misleading validation	3	2	6	Unit, API
R-07 Stock sold twice	3	5	15	Integration transaction tests
R-08 Small-screen layout broken	3	2	6	Mobile viewport Playwright

Gate criteria

All critical journeys pass; no open critical/high unresolved defects; test results documented with reproducible evidence; coverage target **80% line/statements/functions; 70% branches** after the full course suite is written. The starter suite deliberately does **not** enforce that 80% threshold yet.

Test levels

Unit: Vitest. Component: Testing Library. API: Supertest. Integration: in-memory SQLite + Supertest. Browser: Playwright. CI: GitHub Actions.

Environment & data

Local ports 4000 and 5173, Node 20. Each API test opens a fresh in-memory SQLite database. Browser tests register fresh customers. Never manually alter the real database to make a test green.

Entry and exit

Entry: app boots, requirements known, test data available. Exit: critical scenarios pass across required browsers, relevant risks covered, severe defects addressed or release explicitly blocked, reproducible evidence archived.